

Al-Amin Hossain Nahid

Full-Stack Developer | CS Undergraduate

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LinkedIn: al-aminnahid | **GitHub:** AlAminNahid | **Portfolio:** AlAmin

Summary

Computer Science undergraduate at the American International University–Bangladesh (AIUB) with hands-on full-stack engineering experience across web and Android development. Comfortable working end to end — designing relational data models, building and securing REST APIs, and developing the interfaces that consume them. Research interests in Machine Learning, Natural Language Processing, and Large Language Models.

Skills

- **Programming Languages:** C/C++, Java, C#, Python, JavaScript, TypeScript, Kotlin
- **Frontend:** HTML, CSS, Tailwind CSS, React, React Native, Next.js
- **Backend:** NestJS, FastAPI, ASP.NET, PHP
- **Databases & ORMs:** PostgreSQL, MySQL, MongoDB, TypeORM, Prisma, SQLAlchemy
- **Mobile:** Android SDK, Retrofit, MVVM, Material Design
- **AI/LLM:** OpenAI API, LLM Integration, Prompt Engineering
- **Tools:** Git, Docker, Postman, Linux, Android Studio, Firebase, Vercel, Render

Experience

AuraDev LTD | Software Engineering Intern

June 2026 - Present

- Contributing to **GrantOS**, an AI-assisted grant management platform, working end to end across a Next.js front end and a FastAPI backend.
- Developing REST API endpoints and business logic, with relational data models managed through SQLAlchemy and Alembic migrations on PostgreSQL.
- Building client-facing features including document workflows, rich-text editing, a grant pipeline board, and authenticated multi-user access via Clerk.
- Integrating LLM APIs (Anthropic Claude for proposal drafting, Google Gemini for streaming chat) into product features for proposal generation and document analysis.
- **Technologies:** TypeScript, React, Next.js, FastAPI, Python, PostgreSQL, SQLAlchemy, Clerk, Anthropic Claude API, Google Gemini API

Projects

MessMasterApp_Backend | Backend Developer

GitHub

- Built and deployed a modular RESTful API for a smart mess management system using NestJS and PostgreSQL, consumed by an Android client.
- Implemented secure authentication and authorization using JWT access/refresh tokens, Passport.js, bcrypt, and role-based access control.
- Developed core modules for mess management, meal tracking, expenses, utility costs, billing, statistics, and notifications.
- Designed a normalized PostgreSQL schema with TypeORM migrations and class-validator DTOs for data integrity and API validation.
- **Technologies:** TypeScript, NestJS, PostgreSQL, TypeORM, JWT, Passport.js, Docker

MessMaster | Android Developer

GitHub · *Google Play (Internal Testing)*

- Developed a native Android application for shared-living management, covering meal tracking, expense management, utility billing, notices, and member communication.
- Implemented role-based experiences for managers and members, integrating with a RESTful backend using Retrofit and OkHttp with JWT-based authentication.
- Applied MVVM architecture with ViewModel and Kotlin Coroutines for asynchronous operations, with centralized API error handling across repositories.
- Built Material Design interfaces using Kotlin and XML, including dashboards, meal management, utility tracking, notices, profile management, and invite workflows.
- **Technologies:** Kotlin, XML, Android SDK, MVVM, Retrofit, OkHttp, Coroutines, Material Design

Bilimbi Studio | Frontend Developer

Live · GitHub

- Built and deployed a production marketing website for Bilimbi Studio, showcasing its projects, team, development process, and contact functionality.
- Developed a scalable design-token system with Tailwind CSS v4 to maintain consistent branding and styling across the application.
- Structured the website as a content-driven system using typed constants, separating content from UI components for easier maintenance and updates.
- Implemented a Next.js API route for the contact form with Zod-based validation and external email service integration.
- **Technologies:** TypeScript, Next.js, React, Tailwind CSS, Zod

MediCare | Full-Stack Developer

Live · GitHub

- Built a full-stack medical chamber management platform enabling doctors to manage chambers and patients to book serial-based appointments online.
- Developed isolated role-based portals for Admin, Doctor, and Patient with JWT authentication and server-side authorization guards.
- Implemented a serial-queue appointment system with daily serial generation and real-time queue management across Waiting, Serving, and Done states.
- Designed a PostgreSQL data model using TypeORM entities and versioned migrations, with the application deployed across Vercel, Render, and Neon.
- **Technologies:** TypeScript, Next.js, React, Tailwind CSS, NestJS, TypeORM, PostgreSQL, Zod, JWT

AI Chat | Full-Stack AI Developer

Live · GitHub

- Developed a full-stack AI chat application with persistent multi-conversation history using Next.js, Prisma, and PostgreSQL.
- Integrated the Google Gemini API through server-side Next.js route handlers with streamed AI responses, keeping API credentials secure on the server.
- Implemented conversation management with validated REST APIs, typed error handling, and reusable React components and hooks.
- Built a responsive chat interface supporting formatted Markdown responses, including code blocks and lists.
- **Technologies:** TypeScript, Next.js, React, Tailwind CSS, Prisma, PostgreSQL, Google Gemini API

Publications & Research

Accepted Publication

- **Al Amin Hossain Nahid**, Ferdus Hossain, Tariful Islam Fahim, and Salman Zzoha. “Intelligent Waste Classification for Sustainable Urban Development: A MobileNetV3-Based Deep Learning Approach Using the BDWaste Dataset.” To appear in *Proceedings of the International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII 2026)*, 2026.

Ongoing Research

- **Al Amin Hossain Nahid**, Chowdhury Shakil Hasan, Rakibul Islam Akash, Abdullah Al Rafe Ayon. “An AI-Based Legal Advisory System for Traffic Violation Verification Using Official Traffic Regulations.” Manuscript in preparation.

Education

American International University-Bangladesh

B.Sc. in Computer Science and Engineering

Sept 2022 – Present

CGPA: 3.64/4.00

References

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